

OM JOGLEKAR

✉ joglekar.math@gmail.com

🌐 [my website](#)

Examination	Majors	Institute	Year	CPI/%
Graduation*	BS+MSc Dual degree	IIT Bombay	2021–2026	8.42

*ongoing

RESEARCH INTERESTS

I am very curious about all fields of pure mathematics, but specifically I am very enthusiastic about algebraic combinatorics and enumerative combinatorics. I also take interest in algebraic graph theory.

KEY COURSES UNDERTAKEN

Advanced Math Courses	Hyperplane arrangements, Hopf Algebra, Graph Theory, Species and Operads, Topics in Combinatorics, Enumerative Combinatorics, Algebraic Combinatorics, Category Theory, Algebraic Graph Theory, Coding Theory*
Intermediate Math Courses	Combinatorics, Representation Theory of finite groups, Topology, Galois Theory, Fourier Analysis, Functional Analysis, Module and Ring Theory, Lie Theory, Non-Commutative Algebra, Optimisation, Differential Geometry*, Analytic Number Theory*
Computer Science	Computer Programming and Utilization, Discrete Structures, Quantum Computing, Logic in Computer Science
Physics	Quantum Physics and its Applications, Basics of Electromagnetism

** will be completed by April 2025

PROJECTS AND RESEARCH EXPERIENCE

Hopf Algebras and Hyperplane Arrangements

Oct '23 - Aug '25

Self Reading Project | Instructor: Prof Swapneel Mahajan, Dept. of Mathematics, IITB

- Extensively read about Hopf Algebras and linked concepts in Algebraic combinatorics. Explicitly reading from the book **Coxeter Bialgebras** by Swapneel Mahajan and Marcelo Aguiar.
- Learnt about **Hyperplane arrangements** and their connections to Hopf algebras. In particular, learning about category theoretical aspects of Hopf Algebra and developing enough theory to enter into the algebraic geometrical aspects of coxeter algebras.
- In process of making a detailed report and short notes for my reading project and also preparing a video presentation on the same which I shall present.
- Developed combinatorial proofs for multiple identities that seem to have no direct combinatorial proof, but rather a Hopf algebraic proof.

Skolem Sequences

Jun '24 - Aug '24

Research Project – Polymath | Guide: Prof. Saad Mneimneh, Dept. of Computer Science, Hunter College, CUNY

- Investigated novel properties of **Infinite Skolem Sequences**, a relatively unexplored area, and their variations under Professor Mneimneh.
- Worked on connections of such sequences with **Steiner triple systems** and general combinatorial designs. Also explored the connections with Wythoff pairs.

- Played around with variations and slightly modified with the sequence to find connections with **rabbit sequences** and Fibonacci word sequence.
- Worked in a team to develop some algorithms to produce a Skolem sequence and fill the blanks using an arbitrary matrix.

Promoting Undergraduate Mathematics in High School

Jul '22 - Jul '23

Summer Undergraduate Research Project | Guide: Prof. Rekha Santhanam, Dept. of Mathematics, IITB

- Worked in a team of **11 Math UGs** to introduce UG math concepts to the school curriculum.
- Studied the existing curriculum to identify math topics to be introduced in a fun and easy way to high school students who will be exploring probability soon enough.
- Worked on introducing **Probability and game theory** concepts to students in the form of a **novel**.
- **Published** the novel '*The Pirate Ship*'. Its sequel '*A Mathematical Spectacle*' has been approved and is ready to be published as a second book that has higher concepts of probability.

Advanced Algebraic Combinatorics

Oct '24 - present

Self reading project

- Reading up on advanced combinatorial structures including Association schemes and Matroids. Explicitly reading from **Matroid Theory** by J. Oxley and **Algebraic Combinatorics** by Bannai, Ito, Tanaka and Tanaka.
- Simultaneously working through Lattice theory and its applications to poset theory.
- Finished reading **Richard Stanley's** book on Algebraic Combinatorics and am currently going deeper into the commutative algebra related sections.

Theory of Symmetric Functions

Jun '24 - Jan '25

Reading project | Guide: Prof. Arvind Ayyer, Department of Mathematics, IISc Bangalore

- Extensively read **Richard Stanley's** book on Enumerative Combinatorics (Chapter 7) and went deeper into the symmetric function related sections.
- Explored more about the **RSK** algorithm, its symmetry and its other interesting applications. Besides the RSK algorithm, explored more about **Littlewood-Richardson** coefficients.
- Developed an **algorithm** that can be applied by hand to generate the entire poset of Integer Partitions, ordered by the dominance order, of a positive integer.
- Explored the **representation theory** of the symmetric group and deduced a formula for the generating function of simple characters of the symmetric group.

Solvable Lattice Models

Jun '23 - Aug '23

Research Project – Polymath | Guide: Prof. Ben Brubaker, Dept. of Mathematics, Massachusetts Institute of Technology

- Worked on the topic of solvable lattice models under MIT professor Ben Brubaker.
- Worked on finding **partition functions** of various lattice models and figuring out which lattice models generate **symmetric partition functions**.
- Extending the lattice model boundary set conditions to multiple allowable states and working on their partition functions.
- Developed further theory using the **Yang-Baxter equation** to find additional results.

Intro to Combinatorics

May '22 - Jul '22

Reading Project – Summer of Sciences | Guide: Dhawal Singh, Manager, MnP club, IIT Bombay

- Covered lots of theory on Sperner posets and Young's lattice. Was first introduced to combinatorics through the EKR theorem.
- Learnt about core concepts of Combinatorics including Stirling numbers and basic graph theory. Complemented it with basics of Extremal graph theory which included the **Zarankiewicz problem** and the **Erdős–Stone theorem**.
- Made a **video report** on Ferrers' shapes and presented a new idea and a proposition on it.

TECHNICAL SKILLS

Languages	Python, Java, C/C++
Web Technologies	HTML, CSS
Database	SQL
Software	MATLAB, SageMath
Libraries	C/C++: simplecpp, Turbo Python: NumPy, Pandas

POSITIONS OF RESPONSIBILITY

Guide and Teacher of Mathematics | National Service Scheme (NSS) | IIT Bombay Aug '22 - Dec '22

Teaching students of grade 12 and helping them prepare for JEE

- Mentoring and helping **40+** financially constrained students of grade 12 and solving their doubts.
- Helping them clear their concepts and teaching them the finer topics they are facing difficulties in.

Teaching Assistant | Math Department, IIT Bombay

Teaching Assistant for multiple courses in the Math Department

- Served as a Teaching assistant for various first year undergraduate courses in the Mathematics Department including MA109, MA111, MA105, MA106 and MA108.
- Involved conducting **problem solving** tutorial sessions every week for a batch of students.
- Had the opportunity to serve as a head assistant and take part in **setting the question paper** for the midterm and final exams for the students.

DAMP mentor | Math department, IIT Bombay

May '23 - Apr '25

Department Academic Mentorship Program, Mathematics department, IIT Bombay

- **Mentored** sophomores with their academic difficulties and providing credible **counsel** for their personal issues and mental health.
- Spearheaded and executed the **sophomore 101** session as a speaker for the topic of branch change in the session during the year 2023.
- Curated plenty of help sessions and doubt-clearing sessions for my mentees and juniors in some of the hardest courses for that semester.

Summer of Sciences mentor | MnP Club, IIT Bombay

May - Jul ('23, '24, '25)

Mentor for projects listed under the Summer of Science event, Math and Physics club, IITB

- Mentored students over a period of three years in the topics - **Graph theory, Advanced Combinatorics, Linear Algebra** and **Sets and Category theory**.
- Guided my mentees on how to go about doing their project and clearing occasional doubts. A schedule of two sessions a week consisted of a theory lecture followed by a problem solving one.
- Provided relevant **feedback** after reviewing their mid-term and end-term project reports and also evaluated their video submission in order to provide the completion certificate.

Campaign Coordinator | Abhyuday | IIT Bombay

Jan '22 - Jan '23

Largest student-run social body for leadership and impact in India | Patronage: UNICEF, UNESCO

- Worked in a Team of **50+** volunteers to conduct campaigns to bring out a change in the society.
- Headed a group of **20+** volunteers to organize and conduct a **food distribution drive** in collaboration with **Robinhood Army** in the slums of Powai and coordinated logistics for the children.
- Contacted **10+** schools for executing **Career Counselling Campaign** for underprivileged children.
- Took lead and managed the debating event in the **Abhyuday Social Fest**.

EXTRA CURRICULAR ACHIEVEMENTS

- Practised **Taekwondo** for four years and was awarded a **Black Belt** in it by Kukiwon, South Korea.
- Currently pursuing a course in **French** and **Japanese**, which will lead to B1 and N4 level certificates.
- Learnt **Sanskrit** from the Sanskrit Education Cell, IIT Bombay which has tied up with CSU.
- Placed in the top 10 in the **Pan-IIT Cycling** Competition (June 2022), organized by IIT BHU.
- Placed first in an all-India quiz organised by **Presenova** and hosted by quiz master **Pick Brain**.
- Represented Presidency School RT Nagar as a part of its **football team** at inter school matches.
- Participated in and won gold and silver medals in various subject Olympiads at the school level.
- Availed the Scholarship for Higher Education (SHE) under **INSPIRE** for performing in top 1% in Class XII Board Examination and securing an all-India rank of less than 10k in JEE Advanced.
- Completed a 2 year long National Cadet Corps (**NCC**) course to obtain an A grade certificate.
- Completed a 1-year course on health and fitness conducted by **NSO**.
- Participated in an all-Karnataka state level **roller skating** competition.